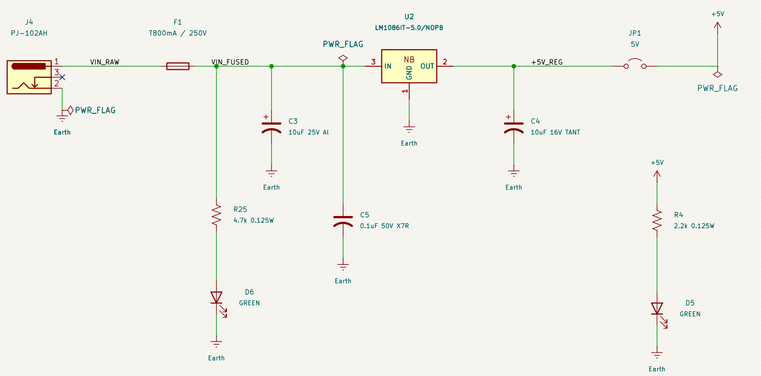
Fill-A-Bot Main Control Board

Power Input and 5V Regulator

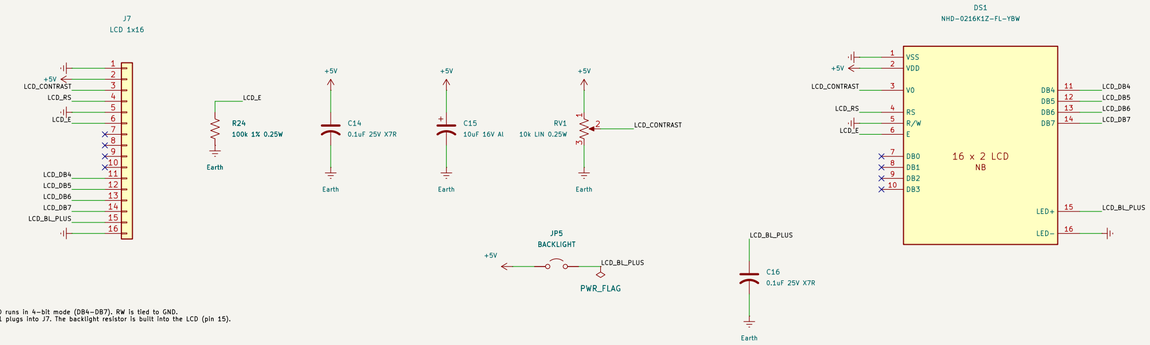


F1 is a 5x20mm fuse in a holder so it can be swapped.

Input is 9–12V DC, center positive.
U2 gets a heat sink (HSE14-38+225P).
D5 and D6 light up when the 5V and 9V rails are on.

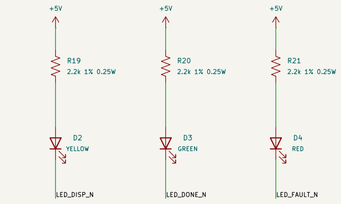


16x2 LCD



LCD runs in 4-bit mode (DB4–DB7), RW is tied to GND.
DS1 plugs into J7. The backlight resistor is built into the LCD (pin 15).

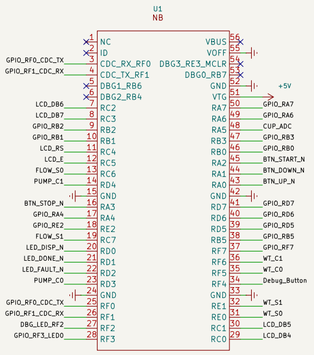
Status LEDs



Curiosity Nano, Extra Headers, Debug

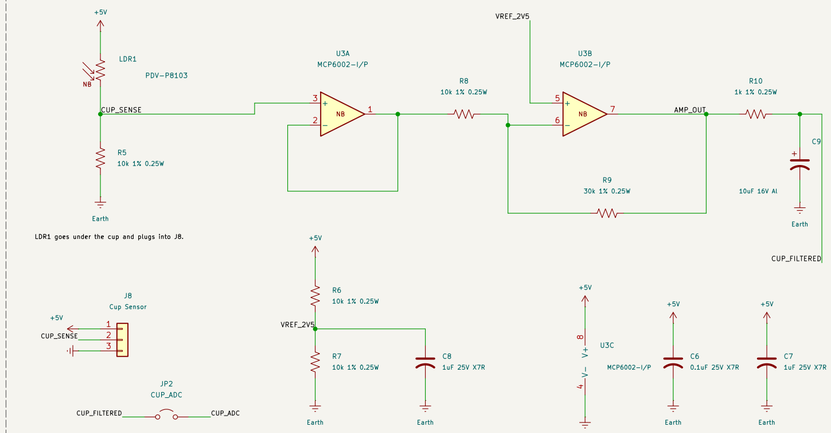
Ribbon input pull-downs (100k): an unplugged cable reads 00





Extra headers for spare pins

Cup Sensor (LDR + Op Amp)



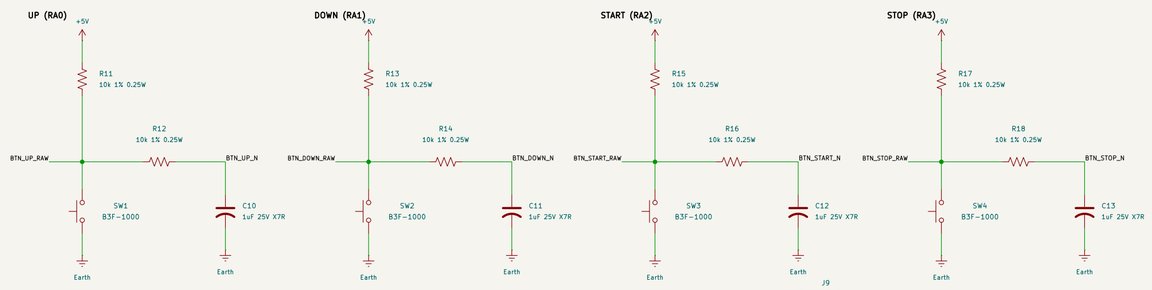
LDR1 goes under the cup and plugs into J8.

U3A buffers the LDR. U3B has a gain of −3 around 2.5V.
R10 and C9 make a low-pass filter (about 16 Hz).

Test Points



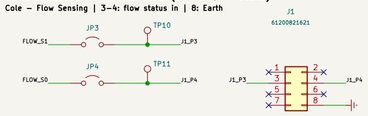
Buttons



Each button has a 10k pull-up and a 10k/1uF RC debounce.
J9 is for buttons mounted off the board.

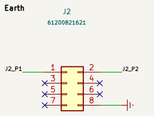
Ribbon Cables to Teammates (2-wire, 4-state)

Cole – Flow Sensing | 3–4: flow status in | 8: Earth



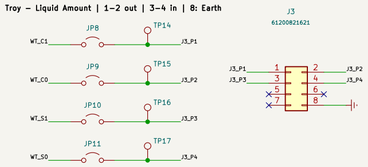
FLOW_S1:S0 = 00 no flow, 01 flowing,
10 not used (no flow), 11 sensor fault

Clay – Pump Control | 1–2: pump command out | 8: Earth



PUMP_C1:C0 = 00 stop, 01 run,
10 and 11 not used (pump stays off)

Troy – Liquid Amount | 1–2 out | 3–4 in | 8: Earth



WT_C1:C0 = preset 1 to 4 (00, 01, 10, 11)
WT_S1:S0 = 00 not ready, 01 below target,
10 target reached, 11 error

Nikita Bangiyev – EGR 304 Team 201

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File: EGR304_Main_Control_Nikita.kicad_sch

Title: Fill-A-Bot Main Control Board

Size: A2	Date: 2026-10-25	**Rev: 3**
KiCad E.D.A. 10.0.7		Id: 1/1